

Math 162

Problem set 3 solutions

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1. Given a partition $P := \{t_0, \dots, t_n\}$ of $[1, x]$, we write $P' := \{t_0y, \dots, t_ny\}$ for the partition of $[y, xy]$ suggested in the hint (note that $t_{i-1}y < t_iy$ for any $1 \leq i \leq n$ because $y > 0$). Since the function $f(t) := \frac{1}{t}$ is decreasing on $[1, x]$, it follows that

$$\inf\{f(t) \mid t_{i-1} \leq t \leq t_i\} = \frac{1}{t_i} \quad \text{and} \quad \sup\{f(t) \mid t_{i-1} \leq t \leq t_i\} = \frac{1}{t_{i-1}}.$$

Similarly, we have

$$\inf\{f(t) \mid t_{i-1}y \leq t \leq t_iy\} = \frac{1}{t_iy} \quad \text{and} \quad \sup\{f(t) \mid t_{i-1}y \leq t \leq t_iy\} = \frac{1}{t_{i-1}y}.$$

It follows that

$$L(f, P') = \sum_{i=1}^n \frac{1}{t_iy} (t_iy - t_{i-1}y) = \sum_{i=1}^n \frac{1}{t_i} (t_i - t_{i-1}) = L(f, P).$$

Similarly, we have $U(f, P') = U(f, P)$.

The continuity of f guarantees that $\int_1^x \frac{1}{t} dt$ and $\int_y^{xy} \frac{1}{t} dt$ both exist. By the definition of the Darboux integral, the integral $\int_1^x \frac{1}{t} dt$ is the unique number satisfying

$$L(f, P) \leq \int_1^x \frac{1}{t} dt \leq U(f, P)$$

for all partitions P of $[1, x]$. But for all such P we have

$$L(f, P) = L(f, P') \leq \int_y^{xy} \frac{1}{t} dt \leq U(f, P') = U(f, P),$$

so these two integrals are equal.

To finish the problem, we deduce that

$$\log x + \log y = \int_1^x \frac{1}{t} dt + \int_1^y \frac{1}{t} dt = \int_y^{xy} \frac{1}{t} dt + \int_1^y \frac{1}{t} dt = \int_1^{xy} \frac{1}{t} dt = \log(xy).$$

2. Since f is continuous, the extreme value theorem says that there exist $x_{\min}, x_{\max} \in [a, b]$ such that $f(x_{\min}) \leq f(x) \leq f(x_{\max})$ for all $x \in [a, b]$. Then the basic bound on integrals implies that

$$f(x_{\min})(b-a) \leq \int_a^b f(x) dx \leq f(x_{\max})(b-a),$$

or equivalently

$$f(x_{\min}) \leq \frac{1}{b-a} \int_a^b f(x) dx \leq f(x_{\max}).$$

Using the continuity of f again, the intermediate value theorem says that there exists c between $x_{\min}$ and $x_{\max}$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx,$$

so we are done.

3. (a) Fix $\epsilon > 0$, and let $\delta := \epsilon^2$. By the triangle inequality, we have

$$|\sqrt{x} - \sqrt{y}| \leq \sqrt{x} + \sqrt{y}$$

for any $x, y \geq 0$, from which we see that

$$|\sqrt{x} - \sqrt{y}|^2 \leq |\sqrt{x} - \sqrt{y}|(|\sqrt{x}| + |\sqrt{y}|) = |\sqrt{x} - \sqrt{y}|(\sqrt{x} + \sqrt{y}) = |x - y|.$$

Thus $|x - y| < \delta$ implies that $|\sqrt{x} - \sqrt{y}|^2 < \delta = \epsilon^2$ and hence $|\sqrt{x} - \sqrt{y}| < \epsilon$ as needed.

- (b) We will use the following properties of $\sin : \mathbb{R} \rightarrow \mathbb{R}$:

- for any $N > 0$, there exists $x > N$ such that $\sin x = 1$;
- $\sin x = 1$ implies that $\sin(x + \pi) = -1$.

The nature of the number π is irrelevant here: in fact, the following example could be made to work with $\sin$ replaced by any nonconstant continuous periodic function.

Consider the function $f : [0, \infty) \rightarrow \mathbb{R}$ given by $f(x) := \sin(x^2)$ (this would work equally well with $[0, \infty)$ replaced by $\mathbb{R}$). It is continuous, being the composition of the continuous functions $\sin$ and $x \mapsto x^2$, and f is bounded because $\sin$ is bounded.

To show that f is not uniformly continuous on $[0, \infty)$, we must show that there exists $\epsilon > 0$ such that for any $\delta > 0$, there exist $x, y \geq 0$ such that $|x - y| < \delta$ and $|f(x) - f(y)| \geq \epsilon$. Let $\epsilon := 1$ and fix $\delta > 0$. Choose $z > \frac{\pi^2}{\delta^2}$ such that $\sin z = 1$, and let $x := \sqrt{z}$, $y := \sqrt{x^2 + \pi}$. Then we have $x > \frac{\pi}{\delta}$, and hence

$$|x - y| = \sqrt{x^2 + \pi} - x = \frac{\pi}{\sqrt{x^2 + \pi} + x} < \frac{\pi}{x} < \delta.$$

Moreover, we have $\sin x^2 = \sin z = 1$, and $\sin y^2 = \sin(x^2 + \pi) = -1$. Thus

$$|\sin(x^2) - \sin(y^2)| = 2 > 1 = \epsilon$$

as needed.

- (c) Let $f : (0, 1] \rightarrow \mathbb{R}$ be defined by $f(x) := \sin(\frac{1}{x})$. This function is continuous because it is the composition of the continuous functions $\sin$ and $x \mapsto \frac{1}{x}$, and f is bounded because $\sin$ is bounded. To prove that f is not uniformly continuous on $(0, 1]$, we again let $\epsilon := 1$ and fix $\delta > 0$. Choose $z > \max(\frac{\pi}{\delta}, 1)$ such that $\sin z = 1$, and let $x := \frac{1}{z}$, $y := \frac{x}{1+x\pi}$. Then we have $x < \min(\frac{\delta}{\pi}, 1)$, and hence

$$|x - y| = x - \frac{x}{1+x\pi} = \frac{x^2\pi}{1+x\pi} < \frac{x\pi}{1+x\pi} < x\pi < \delta.$$

Moreover, we have $\sin \frac{1}{x} = \sin z = 1$ and $\sin \frac{1}{y} = \sin(\frac{1}{x} + \pi) = -1$, so that

$$|\sin \frac{1}{x} - \sin \frac{1}{y}| = 2 > 1 = \epsilon.$$